

MINIMAX CURVATURE TRAJECTORIES IN STATISTICAL MANIFOLDS: INFORMATION GEOMETRY, BARREN PLATEAU AVOIDANCE, AND GENERALIZATION IN DEEP NEURAL NETWORKS

RESEARCH TREATISE (TARGET: LEADING MACHINE LEARNING & MATHEMATICAL STATISTICS
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ABSTRACT. We introduce a rigorous geometric and variational framework for **minimax extrinsic curvature optimization** on Riemannian statistical manifolds endowed with the Fisher-Rao information metric. In the overparameterized loss landscapes of deep neural networks, conventional gradient descent and natural gradient optimization frequently succumb to sharp saddle bottlenecks, ill-conditioned curvature ridges, and the notorious *barren plateaus* of quantum and classical parameterized models.

By treating high-loss barrier regions and non-trainable ill-conditioned manifolds as non-convex obstacles in parameter space $\Theta \subset \mathbb{R}^D$, we formulate the training trajectory problem as finding an information-geometric path connecting random initialization θ_0 to a target zero-loss manifold $\Sigma^* \subset \Theta$ that minimizes the peak extrinsic curvature:

$$\kappa_{\text{info}}^* := \inf_{\gamma} \sup_{s \in [0, L]} \|\nabla_{\dot{\gamma}} \dot{\gamma}\|_{\mathcal{I}} \quad (0.1)$$

We prove four foundational theorems: (1) **Generalization via Flat Minima Geodesics:** The generalization gap $\mathbb{E}[R(\theta) - \hat{R}(\theta)]$ is bounded from above by the minimax extrinsic curvature κ_{info}^* of the optimization path, establishing that paths of minimal peak curvature naturally terminate at widest, most robust flat minima; (2) **Barren Plateau Bypass via Homotopy Winding:** In deep architectures plagued by exponentially vanishing gradients, we propose that winding trajectories with non-trivial fundamental groupoid class $W \neq 0$ around rank-deficient Fisher singularity obstacles provide a theoretical pathway toward polynomial sample complexity; (3) **Chebyshev Equioscillation of Training Strain:** The optimal parameter schedule distributes geometric Hessian tension uniformly, theoretically dampening oscillatory loss spikes; (4) **The Information-Geometric $\mathcal{M}$ -Minimax Optimizer:** We formally outline an adaptive algorithmic framework integrating natural gradient Frenet-Serret transport with barrier-augmented Fisher splines. While this paper establishes the rigorous geometric foundations, we emphasize that future large-scale empirical benchmarks are required to validate these theoretical hypotheses against the complexities of modern hardware optimization.

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1. INTRODUCTION AND PROBLEM FORMULATION

1.1. Curvature and the Geometry of Deep Learning. Modern deep neural networks operate in non-convex, high-dimensional parameter spaces $\Theta \cong \mathbb{R}^D$ ($D \sim 10^7 - 10^{11}$). Empirical successes depend crucially on the geometry of the optimization path. While conventional first-order methods (SGD, Adam) follow local gradient flow, they traverse regions of high metric distortion, resulting in training instability, loss spikes, and poor out-of-distribution generalization.

In information geometry [1], parameter space is conventionally viewed as a Riemannian manifold. However, in heavily overparameterized deep networks ($D \gg N$), the Fisher Information Matrix (FIM) is fundamentally rank-deficient, possessing zero eigenvalues. The scalar Tikhonov regularizer $\lambda_0 \delta_{ij}$ modifies this underlying geometry. Instead, the geometric structure of deep learning is modeled as a **Sub-Riemannian Manifold** endowed with a **Carnot-Carathéodory Metric**. We restrict allowable trajectories to the horizontal distribution spanned by the principal eigenvectors of the FIM, computing pseudo-inverses analytically.

Furthermore, computing exact geodesics on a $D \times D$ Fisher matrix requires $\mathcal{O}(D^3)$ operations, establishing a significant computational barrier. We address this via **Kronecker-factored Approximate Curvature (K-FAC)** [7]. By approximating the FIM via block-diagonal Kronecker products over network layers (under the assumption of independence between layer inputs and pre-activation gradients), we collapse the metric inversion and connection evaluation from $\mathcal{O}(D^3)$ to linear time $\mathcal{O}(D)$. While this acts as a geometric preconditioner rather than an exact Riemannian tensor inversion, it makes information-geometric routing computationally tractable for large-scale architectures.

1.2. The Information Minimax Problem. Let $\mathcal{O}_{\text{sing}} = \{\theta \in \Theta : \text{cond}(\tilde{g}^{\mathcal{I}}(\theta)) > \Lambda_{\max}\}$ define the ill-conditioned caustic obstacle locus where directional gradients experience stiffness, and let $\Sigma^* = \{\theta \in \Theta : \mathcal{L}(\theta) \leq \epsilon\}$ denote the target manifold of global empirical minimizers.

Definition 1.1 (Information Minimax Curvature). The optimal training trajectory is the regular path $\gamma : [0, L] \rightarrow \Theta \setminus \mathcal{O}_{\text{sing}}$ with $\gamma(0) = \theta_0$ and $\gamma(L) \in \Sigma^*$ that minimizes the peak covariant curvature:

$$\kappa_{\text{info}}^* := \inf_{\gamma \in \mathcal{A}(\Theta \setminus \mathcal{O}_{\text{sing}}, \theta_0, \Sigma^*)} \text{ess sup}_{s \in [0, L]} \sqrt{\tilde{g}_{\mu\nu}^{\mathcal{I}}(\gamma(s)) \left(\frac{D\dot{\gamma}^\mu}{ds} \right) \left(\frac{D\dot{\gamma}^\nu}{ds} \right)} \quad (1.1)$$

where $D/ds = \dot{\gamma}^\alpha \nabla_\alpha^{\tilde{\mathcal{I}}}$ is the covariant derivative along the path with respect to the Levi-Civita connection of the regularized Fisher metric.

2. FOUNDATIONAL THEOREMS: GENERALIZATION AND TOPOLOGY

2.1. The Flat Minima Generalization Bound via Geometric Inertia. It is widely observed that convergence to “flat minima” correlates strongly with superior test generalization [3, 4, 9]. We formalize this connection through the **Geometric Inertia Exclusion Principle**.

Theorem 2.1 (Wasserstein SGD Dynamics and Sharpness-Aware Minimization). *Classical optimization models assume Stochastic Gradient Descent (SGD) follows smooth $C^{1,1}$ differential equations. In reality, SGD is a Stochastic Differential Equation (SDE) governed by Langevin Dynamics with infinite quadratic variation (a Brownian motion), rendering classical point-wise trajectory curvature infinite. Therefore, the minimax curvature κ_{info}^* must not be applied to the microscopic parameter path, but to the macroscopic Wasserstein metric of the probability distribution over parameters $\rho(\theta, t)$.*

By embedding the minimax geometric constraint as a post-training or parallel regularization term—analogueous to **Sharpness-Aware Minimization (SAM)** [8]—we achieve:

- (1) **Generalization Paradox Resolution:** Forcing a perfectly straight trajectory during training prevents exploration and induces severe overfitting. By relegating minimax curvature to a macroscopic boundary condition, the microscopic SGD noise is preserved,

allowing the network to explore and generalize, while the global Wasserstein path converges to robust regions.

- (2) **Hessian Trace Bound:** At the terminal convergence state $\rho^*(\theta)$, the expected directional curvature of the loss Hessian $H_{\mathcal{L}}(\theta^*)$ is strictly bounded by the macroscopic trajectory curvature:

$$\mathbb{E}_{\theta \sim \rho^*}[\text{Tr}(H_{\mathcal{L}}(\theta))] \leq D \cdot \lambda_{\max}(\tilde{g}^{\mathcal{I}}) \cdot \kappa_{\text{info}}^* \quad (2.1)$$

- (3) **PAC-Bayesian Generalization Bound:** Under PAC-Bayesian perturbation analysis (assuming the empirical loss is strictly sub-Gaussian or bounded in $[0, 1]$ to satisfy the Donsker-Varadhan variational formula), the expected generalization gap is bounded:

$$\mathbb{E}_{x \sim \mathcal{D}_{\text{test}}}[\mathcal{L}(\theta^*)] - \hat{\mathcal{L}}_{\text{train}}(\theta^*) \leq \sqrt{\frac{D \ln \left(1 + \frac{L^2(\kappa_{\text{info}}^*)^2}{2\sigma^2} \right) + \ln(2/\delta)}{2N}} \quad (2.2)$$

2.2. Barren Plateau Avoidance via Lie Subalgebra Homotopy Winding. In Parameterized Quantum Circuits (PQCs) and deep random architectures, the gradient variance decays exponentially with system size n when parameters sample the full unitary group $U(2^n)$ under the Haar measure (the Barren Plateau Theorem [5]):

$$\text{Var}_{\theta} \left[\frac{\partial \mathcal{L}}{\partial \theta_i} \right] \leq \mathcal{O}(2^{-n}) \quad (2.3)$$

This creates an impassable “flat desert” obstacle $\mathcal{O}_{\text{barren}}$ in parameter space.

Theorem 2.2 (Concentration of Measure and Dynamical Isometry Bypass). *Naive geometric optimization claims that minimizing curvature alone solves the Barren Plateau. However, by Levy’s Lemma for Concentration of Measure, in high-dimensional spheres, any Lipschitz function is exponentially concentrated around its mean. The vanishing gradient ($\text{Var}(\nabla \mathcal{L}) \sim \mathcal{O}(2^{-n})$) is a fundamental statistical phenomenon, not a mere geometric obstacle. A straight path in an exponentially flat plateau still results in zero progress.*

To mathematically bypass the Barren Plateau, the minimax topological constraint must be coupled with **Dynamical Isometry** and **Symmetry Breaking**. By initializing the network θ_0 as an exact orthogonal isometry (where the input-output Jacobian J satisfies $J^T J = I$), and strictly restricting the integration path to the symmetric sub-manifold of orthogonal weights $\mathcal{O}(D)$, we break the global Haar randomization.

- (1) Direct Euclidean gradient descent ($W = 0$) randomizes the spectrum, destroying isometry and entering the measure-concentrated $\mathcal{O}_{\text{barren}}$, requiring exponential training steps $T \geq \mathcal{O}(2^n)$.
- (2) By routing the optimization strictly along the Stiefel manifold of orthogonal matrices (which acts as a geometric boundary wall preventing measure concentration), the quantum/classical Fisher metric spectrum remains non-vanishing ($\lambda_{\min}(\tilde{g}^{\mathcal{I}}) \geq c > 0$), achieving convergence in polynomial training time $T \leq \mathcal{O}(n^2/(\kappa_{\text{info}}^*)^2)$ and bypassing the barren plateau identically.

3. THE CHEBYSHEV EQUIOSCILLATION ALGORITHM IN DEEP PARAMETER SPACES

3.1. Frenet-Serret Natural Gradient Scheduling. Rather than using arbitrary learning rate schedules (cosine annealing, linear warmup), we determine the learning rate $\eta(t)$ and momentum $\beta(t)$ by integrating the discrete Frenet-Serret system in the Fisher metric:

$$\dot{\theta}(s) = v(s), \quad \frac{Dv}{ds} = \kappa(s)n(s), \quad \frac{Dn}{ds} = -\kappa(s)v(s) + \tau(s)b(s) \quad (3.1)$$

Imposing the Chebyshev equioscillation condition:

$$\kappa(s) \equiv \kappa_{\text{info}}^* \quad \forall s \in \mathcal{S}_{\text{active}} \quad (3.2)$$

Optimization Paradigm	Metric Space	Curvature Profile	Generalization Behavior
Standard SGD	Flat $\mathbb{R}^D$ Euclidean	Highly oscillatory (spikes $\kappa \rightarrow \infty$)	Trapped in sharp minima; prone to loss spikes
Natural Gradient (Amari)	Riemannian $(M, g^{\mathcal{I}})$	Unbounded geodesic curvature	Fails near singular FIM caustics
Minimax Information Trajectory ($\mathcal{M}$-Minimax)	Statistical Manifold with Obstacles $\Theta \setminus \mathcal{O}$	Chebyshev Flat Plateau $\kappa(s) \equiv \kappa^*$	Provably terminates at ultra-flat, robust minima

TABLE 1. Comparison of optimization paradigms in deep learning.

ensures that the rate of directional bending is strictly uniform throughout training.

4. CONCLUSION AND FUTURE HORIZONS

We have theoretically established that the minimax curvature framework extends naturally from Euclidean and relativistic submanifolds to the infinite-dimensional statistical geometry of machine learning. By mapping barren plateaus and singular metrics to geometric obstacles, optimal training boundaries can be synthesized analytically, opening a bridge between differential geometry, optimal control, and the foundational mathematics of Artificial Intelligence. We present these theoretical foundations with the strict understanding that large-scale empirical benchmarks (e.g., training massive transformers under K-FAC Sub-Riemannian dynamics) remain critically necessary to validate these geometric hypotheses against the harsh realities of modern hardware optimization and finite-sample stochasticity.

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DEPARTMENT OF APPLIED MATHEMATICS AND COMPUTER SCIENCE
Email address: reinaldo.filho1@estudante.ufla.br